

# Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo + Discrepancy

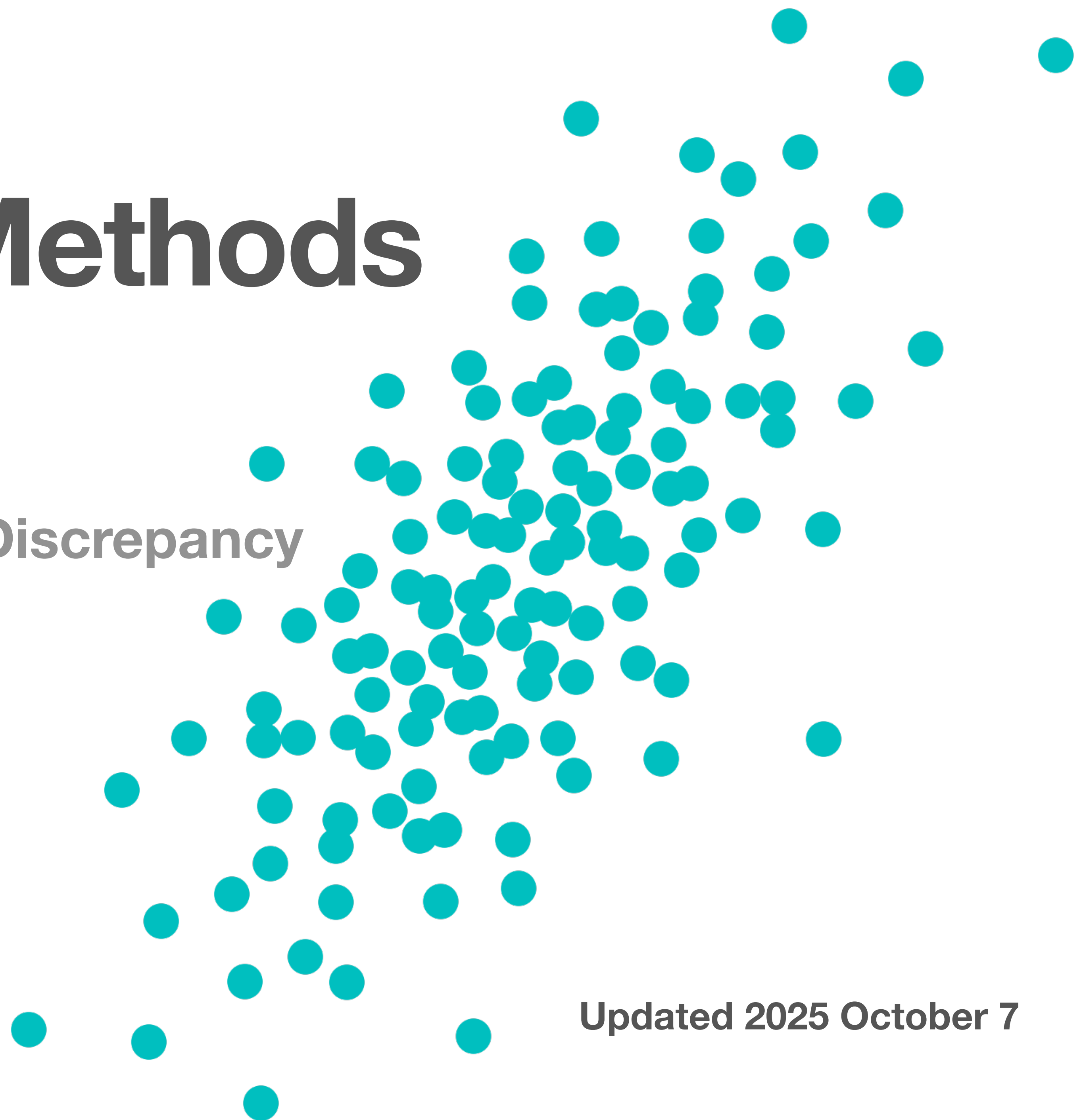
Improving Efficiency

Selected Topics

Git website and repository  
Canvas

Fred Hickernell, Fall 2025

Updated 2025 October 7

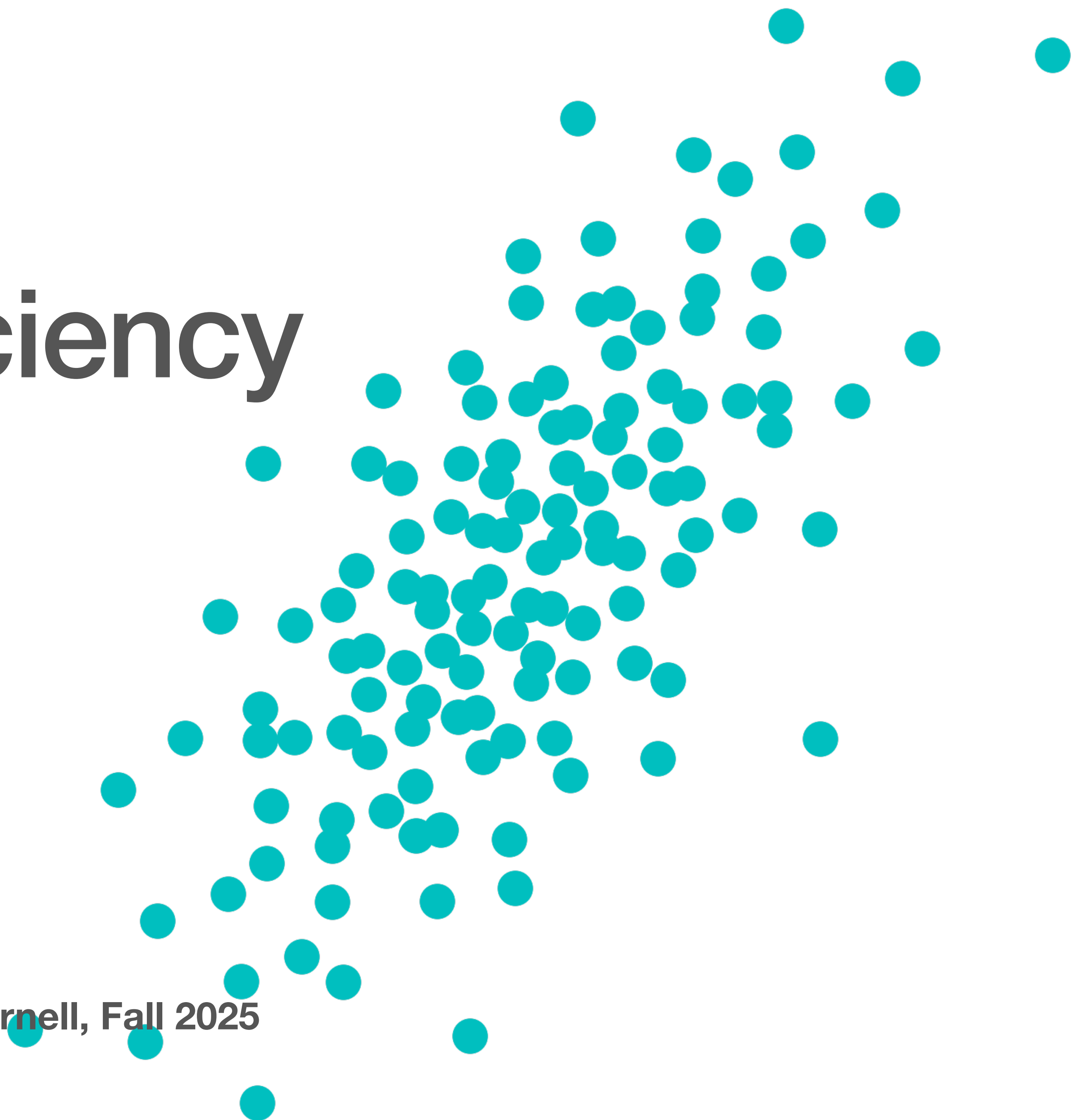


# Improving Efficiency

Owen, Chapters 11– 12

Test 2 Oct 29

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025





# Variance Reduction

$$\mu = \mathbb{E}[f(X)] = \mathbb{E}[g(X)], \quad X \sim \mathcal{U}[0,1]^d$$

Is it better to use

$$\hat{\mu}_{f,n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \quad \text{or} \quad \hat{\mu}_{g,n} = \frac{1}{n} \sum_{i=0}^{n-1} g(X_i), \quad X_0, X_1, \dots \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}[0,1]^d$$

to approximate  $\mu$ ? Since both sample means are unbiased, we can compare their variances (mean squared errors):

$$\text{var}(\hat{\mu}_{f,n}) = \frac{\text{var}(f(X))}{n} \quad \text{var}(\hat{\mu}_{g,n}) = \frac{\text{var}(g(X))}{n}$$

so it depends on the variances of the two functions





# Keister example

$$\mu = \int_{\mathbb{R}^d} \cos(\|t\|) \exp(-\|t\|^2) dt$$

$$\mathbf{x} = (\Phi(at_1), \dots, \Phi(at_d)), \quad \Phi \text{ is the CDF of the standard normal}$$

$$d\mathbf{x} = a^d \phi(at_1) \cdots \phi(at_d) dt = \frac{a^d \exp(-a^2 \|t\|^2/2)}{(2\pi)^{d/2}} dt$$

$$t = (\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d)) / a$$

$$\mu = \int_{[0,1]^d} \frac{(2\pi)^{d/2}}{a^d} \cos\left(\left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\| / a\right)$$

$$\times \exp\left(-\frac{2-a^2}{2a^2} \left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\|^2\right) d\mathbf{x}, \quad (a^2 \leq 2)$$

Whole family  $f(\mathbf{x}; a)$  for which  $\mu = \mathbb{E}[f(X; a)]$  for  $X \sim \mathcal{U}[0,1]^d$



# Importance sampling aka variable transformations